Project Proposal

Job Tracker: Job Application Management System

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Abstract

Job Tracker is a full-stack web application developed to help job seekers manage and track their job applications efficiently throughout the hiring process. Built using the MERN stack (MongoDB, Express.js, React.js, and Node.js), the system provides a centralized platform for organizing applications, tracking application status, and managing follow-up reminders. The project aims to reduce missed opportunities and improve job search efficiency by offering a structured and user-friendly solution. The system follows the Waterfall Methodology to ensure systematic development and documentation.

Background of the Project

In today's competitive job market, job seekers often apply to multiple companies simultaneously. Managing application details manually through emails, spreadsheets, or notes becomes inefficient and confusing. With advancements in web technologies, it is now possible to build centralized platforms that automate tracking, reminders, and analytics. Job Tracker is designed to address these challenges by providing a modern digital solution for job application management.

Problem Statement

Job seekers face several difficulties in managing their job applications. They often forget where they have applied, fail to track application status updates, and miss follow-up opportunities. There is no centralized view of job search progress, leading to duplicate applications and missed chances. These issues result in an inefficient and unorganized job search process.

Solution Overview

The proposed Job Tracker system provides a centralized platform for managing job applications. It allows users to register securely, add and manage job applications, track application status through various stages, and set automated follow-up reminders. The system includes a dashboard with analytics to help users monitor their job search progress efficiently.

Objective of the project

General Objective:
To develop a web-based job application tracking system using the MERN stack.

Specific Objectives:

- To provide secure user authentication.
- To manage job applications through CRUD operations.
- To track application status efficiently.
- To automate follow-up reminders.
- To provide analytics and insights through a dashboard.

Scope of the project

The scope of the project includes the development of a full-stack web application for job seekers. The system supports user authentication, application tracking, analytics, and reminders. It is designed to be responsive and scalable. Future scope includes mobile app development, email notifications, and AI-powered job recommendations.

Methodology

The Waterfall Methodology is adopted for this project. The development process includes requirement analysis, system design, implementation, integration, testing, and deployment. Each phase is completed sequentially to ensure quality and proper documentation.

Steps of Methodology:

- **Requirement Analysis and Definition**
- **System Software Design**
- **Implementation and Unit Testing**
- **Integration and System Testing**
- **Operation and Maintenance**

"Waterfall methodology" is phase developing methodology. Each phase is documented briefly and sends to the user for approval, if it is approved then the next phase is started. If the first phase is not approved by client or it does not according to the requirement of the client it is re-documented.

Because my project requires the same situation so "Waterfall Methodology" is best to use.

Development Tools

Activities	Tools/Programs
Frontend development	React.js
Backend development	Node.js + Express.js
Database	MongoDB
Platform	Windows
Browsers	Google chrome
Documentation	MS Word
Diagram	Draw.io

Table1

Expected Outcomes

- A fully functional job application tracking system.
- Secure authentication and data protection.
- Organized and efficient job application management.
- Automated reminders for follow-ups.
- Scalable architecture for future enhancements.

Conclusion

Job Tracker provides an effective solution to a real-world problem faced by job seekers. By centralizing job application data and automating tracking and reminders, the system improves efficiency and organization. The project demonstrates strong full-stack development skills using modern web technologies.